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Protocol of a systematic review and network meta-analysis for the prevention and treatment of perinatal depression.

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Title: Protocol of a systematic review and network meta-analysis for the prevention and treatment of perinatal depression.

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Abstract

Introduction

Perinatal depression is common and can often lead to adverse health outcomes for mother and child. Multiple pharmacological and non-pharmacological treatments have been evaluated in meta-analyses for preventing and treating perinatal depression compared to usual care or placebo controls. It is not yet established which of these candidate treatments might be the optimal approach for prevention and treatment.

Methods and analysis

A systematic review and Bayesian network meta-analyses will be conducted. Eight electronic databases shall be searched for randomised controlled trials that have evaluated the effectiveness of treatments for prevention and/or treatment of perinatal depression. Screening of articles shall be conducted by two reviewers independently. One network meta-analysis shall evaluate effectiveness of interventions in preventing depression during the perinatal period, and a second network meta-analysis shall compare the effectiveness of treatments for depression symptoms in women with perinatal depression. Bayesian 95% credible intervals shall be used when comparing pooled mean effect size of each treatment and surface under cumulative ranking area (SCURA) will be used to estimate ranking of treatments' effectiveness.

Ethics and dissemination

We shall report our findings so that healthcare providers can make informed decisions on what might be the optimal approach for addressing perinatal depression to prevent cases and

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improve outcomes in those suffering with depression through knowledge exchange workshops, international conference presentation and journal article publication.

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Strengths and limitations of this study

- This planned systematic review and network meta-analysis shall evaluate all available evidence from randomised controlled trials to evaluate interventions for prevention and treatment of perinatal depression
- Health care providers will be able to compare interventions to each other as well as usual care to judge the optimal approach for treating and preventing depression among perinatal women
- This study will not evaluate how multiple interventions interact with each other

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Introduction

Background

Depression is the leading cause of disability worldwide, it is a major contributor to the global burden of disease, affecting a variety of populations[1]. Depression experienced during pregnancy and 4 weeks after birth, also known as perinatal depression, is common and can affect up to 20% of pregnant mothers[2]. Perinatal depression can lead to a range of adverse health outcomes for the women and the development of their children. Depression during pregnancy can lead to multiple problems including premature delivery, gastrointestinal pain, poorer self-report health and functioning and increased risk of smoking or alcohol abuse[3 4]. Longer-term depression beyond the perinatal stage can lead to later problems during parenting including lower interaction and sensitivity between mother and infant. Long lasting depression has been shown to lead to difficulties in later years within the offspring including emotional and behavioural difficulties[5-7]. Prevention of postnatal depression occurring can be achieved. Identifying those with depression early; either during the antenatal period (during pregnancy) or early in the postnatal period (up to 4 weeks postpartum) provides a critical opportunity for earlier treatment and prevention of poorer outcomes associated with longer lasting postnatal depression [8 9].

Despite its significant burden on maternal and child health, less than half of pregnant women suffering from depression are identified within healthcare[10]. Attitudes towards perinatal depression case-finding among clinicians are positive, but there is a need for support strategies to identify and treat those at risk of perinatal depression within routine practice[11]. A systematic review suggested that the Whooley questions, a set of 2-item yes/no answered

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questions were a valid and feasible approach for identifying possible positive cases of perinatal depression[12].

Once identified, healthcare services can provide interventions for preventing those at risk of depression occurring in the future or, for those with depression, offer treatments. There is a wide variety of intervention that have been investigated using meta-analysis of randomised controlled trials for treating depression symptoms in perinatal women: psychological interventions, pharmacological interventions or combinations of both[13]; psychoeducation or parenting education[14]; psychosocial interventions for treatment and prevention[15-18]; systemically oriented psychotherapies[19]; mindfulness[20]; family therapy[21]; physical activity[22]; and yoga-based interventions[23]. In later stage postnatal women, meta-analyses have suggested cognitive behavioural therapy, interpersonal therapy, counselling and other psychological interventions are effective in treating depression symptoms when compared to usual care[24]. Another meta-analysis on antidepressants for postnatal depression in a small number of studies show that selective serotonin reuptake inhibitors (SSRI) are effective for depression compared to placebo[25].

Clinical guidelines recommend screening and treatment for perinatal depression. However, they do not provide recommendations on which treatments are most effective[26]. Treatment options for depression during pregnancy vary depending on different severities of depression[8]. Many treatments evaluated were identified as having a positive effectiveness on depression symptoms compared to usual care or placebo, but the relative comparability of these treatment has not previously been investigated. Relative comparisons of different treatments would allow healthcare providers to make informed decisions on how different active treatments can be compared. The relative comparison of treatments could also provide

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evidence for the optimal approach to treating perinatal depression based on all available evidence.

Rationale

Using a Bayesian network meta-analysis facilitates for all interventions to be compared equally with one-another by using the direct evidence (within study comparisons of treatments) and indirect evidence (comparing treatments across different studies), which previous systematic review studies and clinical guidelines have not yet explored. This approach can provide evidence for the relative comparative effectiveness of each treatment and the potentially optimal approach for preventing cases and treatment of symptoms for perinatal depression.

We will conduct a comprehensive systematic review of available peer reviewed published evidence for all interventions: pharmacological and non-pharmacological, addressing perinatal depression by conducting a network meta-analysis. Based on this we will be able to compare each treatments' effectiveness with one-another and recommend types of interventions that may optimally address the prevention and treatment of perinatal depression.

Objectives

This study will assess the clinical benefits of different interventions for addressing the prevention and treatment of perinatal depression. Two objectives have been developed for this study.

1.) To identify the optimal approach for preventing perinatal depression in women.

2.) To identify the optimal approach for the treatment of perinatal depression in women.

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Methods

The protocol of this study has been developed following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Protocols guidelines and the PRISMA extension statement on the reporting of systematic reviews that incorporate network meta-analysis of health care interventions[27 28].

Eligibility criteria

Study selection and eligibility criteria was based on the PICOS in relation to objectives 1 and 2 (**table 1.**). Briefly, studies on participants who are perinatal women between 20 weeks gestation to 4 weeks after birth will be included. We selected 4 weeks after birth to include studies with study samples that crossover the antenatal stage and early postnatal stage beyond the two weeks to distinguish perinatal depression "and postpartum blues" [29]. We did not specify any limitations on interventions as this review aims to identify and evaluate all intervention types that address depression within the target population. Given the advantages of the network meta-analysis approach, no limitations will be placed on the comparison group for studies. Outcomes for Objective 1 will be: confirmed cases of depression and for Objective 2: measurements of depression severity or symptoms. Study design will only include randomised controlled trials to minimise the risk of bias when comparing effectiveness of interventions. Eligibility is displayed in table 2. Studies that include participants with substance abuse, psychotic or developmental disorders or medical conditions, in long term care, residential facilities or those in institutions (psychiatric inpatients) were excluded as the treatment needs of these populations' depression symptoms differ compared to those with depression alone[30].

Sources of information

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Databases searched electronically will be MEDLINE, British Nursing Index (BNI), EMBASE, Cumulative Index to Nursing & Allied Health (CINAHL Plus), PsycINFO, Cochrane Central Register of Controlled Trials (CENTRAL), Cochrane Database of Systematic Reviews (CDSR) and Web of Science (WoS). World Health Organization's trials portal and clinicaltrials.gov will be searched to identify unpublished studies or studies still ongoing.

We shall conduct searches of reference lists and forward citation of identified and included studies using the Web of Science database for additional papers. We shall exclude studies that are published systematic review or literature review identified during our electronic searches but shall examine the reference lists for additional candidate studies.

Search Strategy

Searches of online databases will commence in January 2021. The search strategy has been developed based on the two sets of PICOS with one for each review question. We identified search terms taken from previously published meta-analyses on the prevention or treatment of perinatal depression[13-23]. An example full list of search terms for the MEDLINE database can be found in **appendix 1**.

Data management

Studies captured in our search strategy shall be downloaded and stored in EndNote (X9) where duplicates across difference sources will be removed.

Study selection process

Screening of studies shall be conducted using the inclusion and exclusion criteria (**table 2)**. Study selection shall be conducted by two of the reviewers (RS, JG, SGH or IHP), with both reviewers reviewing each retrieved article independently. Differences in assessment or any

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disagreements over the eligibility of studies were resolved by discussion, and in cases of disagreement, the third reviewer (KL) would be consulted. Screening of studies shall take place at two different stages; 1.) title and abstract, where studies will only be excluded if there is clear disparity to eligibility criteria, if it is unclear then articles or included articles will be further screened in stage 2.) full article screening, where eligibility can be decided based on all reported article information, including supplementary materials. Rationale for exclusion of studies will be recorded. Piloting of the data selection process shall take place prior to the full study selection, with 100 articles randomly selected from the search retrieved. Adjustments to the inclusion and exclusion criteria and study selection process maybe made following piloting.

Data extraction

Data extraction shall be conducted by two of the reviewers (RS, JG, SGH or IHP) independently using a standardised extraction form. Differences in assessment or any disagreements for data extraction of studies will be resolved by discussion, and in cases of disagreement, the third reviewer (KL) will be consulted. The data extraction forms were taken from the Cochrane Consumers and Communication Review Group's Data Extraction Template for Cochrane Reviews, and were modified to fit this systematic review. Extracted information will include study setting; study population and participant demographics and baseline characteristics; details of the intervention and control conditions; study methodology; recruitment and study completion rates; outcomes and times of measurement; indicators of acceptability to users; suggested mechanisms of intervention action; information for assessment of the risk of bias. Missing data will be requested from study authors.

Risk of bias assessment

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Risk of bias will be assessed using the version 2 of the Cochrane Risk of Bias tool (RoB2)[31]. This tool includes 5-domains based on risk of biases within the randomisation process, deviation from intervention design, missing outcome data, measurement of outcome and reporting of results. The risk of bias assessment shall be completed following the guidance released with the RoB2. Possible risk of bias judgments shall include 1) low risk, 2) some concerns and 3) high risk of bias. Results of the risk of bias shall be presented within each domain of risk of bias as well as an overall score for each study. Overall risk of bias shall be judged based on the suggested criteria of the RoB2 guidance document. Piloting and adaption of the wording in the risk of bias assessment, if necessary, shall be conducted prior to the full review. Risk of bias will be evaluated by two of the reviewers (RS, JG, SGH or IHP) independently. Differences in assessment or any disagreements will be resolved by discussion, and in cases of disagreement, the third reviewer (KL) will be consulted.

Data analysis

Effect measurements

Data from each study shall be extracted with effect size calculated. For studies addressing review question 1 the number of depression cases after intervention shall be extracted for each arm and the study's sample size shall be extracted allowing for relative risk (RR) to be calculated in each arms' comparison, with RR of less than 1 representing reduced risk of depression. For review question 2 treatment effect shall be calculated using mean difference (MD) if possible, or standardised mean difference (SMD) for depression severity. This is the difference in score changes (from baseline to follow-up) between the two groups divided by the pooled standard deviation (SD) of change; in studies with three or more arms a reference group shall be taken to calculate SMD across arms, with negative values representing reduced

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depression severity. Score changes will be used to control for possible baseline differences between study arms. For studies with multiple follow-up time points, we shall use the longest duration, up to a maximum of 1 year from end of intervention. In studies using median and interquartile ranges we shall impute these following the Cochrane Handbook [32]. Studies which do not report the SD of change from baseline will be imputed following the Cochrane Handbook. A correlation coefficient for imputation of SD of change shall be estimated based on the mean correlation in studies that do report all relevant data. If no studies report baseline, follow-up, and change values we shall take the conservative value of r=0.5 to estimate SD of change. Where possible we shall use the intention to treat sample for analyses. Interventions will be grouped for the network analysis using categories used by previous individual meta-analysis[14-23] as a framework, new emerging interventions not previously evaluated in meta-analysis shall be organised and grouped by agreement with the reviewing team.

Network meta-analysis implementation

Two network meta-analyses shall be conducted for depression prevention, using RR and treatment of depression severity, using SMD. We estimated model consistency by comparing the RR or SMD of the direct (within study comparisons) and indirect comparisons (between study comparisons), where direct comparisons are possible. The network meta-analysis will be conducted with a Bayesian Markov chain Monte Carlo (MCMC) method fitted using the Just Another Gibbs Samplers (JAGS) software within the R Statistical Software using the BUGSnet package (R Core Team. 2020). We shall run four MCMC chains simultaneously in our model and construct two separate MCMC simulations to compare convergence. The Bayesian model shall run 5000 burn in iterations and 100,000 simulation iterations. Convergence shall

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be assessed using potential scale reduction factor (PSRF), where we expect the model reduce to below 1.05. Heterogeneity (direct evidence) and model consistency (direct versus indirect) will be assessed using the node split function of the BUGSsnet package; sources of heterogeneity will be explored between studies. All results of each possible comparison shall be made using RR or SMD and 95% credible intervals (Crls), which can be considered Bayesian equivalent of confidence intervals. Rank probabilities will represent the probability of the ranking performance of each intervention type. Surface under the cumulative ranking (SUCRA) score will also be used to estimate the likelihood of the most effective intervention[33 34]. A limitation of the network meta-analysis approach is that treatments that have not been previously combined in trials cannot be combined in the meta-analysis. This precludes an investigation of whether or not combining two or more treatments provides any extra benefits compared with one treatment only.

Treatments for women with depression during or after pregnancy can vary in different populations and across disease severity[8]. To address this, network meta-regression and subgroup analysis shall be conducted to evaluate study characteristics that may influence the effect sizes of interventions within the network. Factors for exploring in the meta-regressions shall include year of publication, study sample age, whether the study participants were at antenatal or early postnatal stage of motherhood, study sample's baseline depression severity (if possible), risk of bias in all 5 domains and overall risk of bias. Publication bias shall be assessed using two funnel plots of all included studies for each objective; trim and fill analysis shall also be conducted.

Patient involvement

Patients and the public were not involved in the design of the protocol or analysis of this study.

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Ethics and dissemination

Ethics is not required for this study, given that this is a protocol for a systematic review, which utilises published data. The results of the review would be widely disseminated locally, nationally, and internationally. A paper would be submitted to a leading peer-review journal in this field, reporting of the study will adhere to the PRISMA extension statement on the reporting systematic reviews that incorporate network meta-analysis of health care interventions[28]. When presenting our findings from this study the GRADE scoring for evidence and strength of recommendations shall be made following the criteria in the GRADE handbook[35]. The findings shall also be presented at a relevant international conference.

Reference List

1. Friedrich M. Depression is the leading cause of disability around the world. *Jama* 2017;317(15):1517-17.
2. Woody C, Ferrari A, Siskind D, et al. A systematic review and meta-regression of the prevalence and incidence of perinatal depression. *Journal of affective disorders* 2017;219:86-92.
3. Marcus SM. Depression during pregnancy: rates, risks and consequences. *Journal of Population Therapeutics and Clinical Pharmacology* 2009;16(1)
4. Grigoriadis S, VonderPorten EH, Mamisashvili L, et al. The impact of maternal depression during pregnancy on perinatal outcomes: a systematic review and meta-analysis. *The Journal of clinical psychiatry* 2013;74(4):321-41.
5. Murray L, Arteche A, Fearon P, et al. Maternal postnatal depression and the development of depression in offspring up to 16 years of age. *Journal of the American Academy of Child & Adolescent Psychiatry* 2011;50(5):460-70.
6. Velders FP, Dieleman G, Henrichs J, et al. Prenatal and postnatal psychological symptoms of parents and family functioning: the impact on child emotional and behavioural problems. *European child & adolescent psychiatry* 2011;20(7):341-50.
7. Kersten-Alvarez LE, Hosman CM, Riksen-Walraven JM, et al. Early school outcomes for children of postpartum depressed mothers: comparison with a community sample. *Child Psychiatry & Human Development* 2012;43(2):201-18.
8. Vigod SN, Wilson CA, Howard LM. Depression in pregnancy. *Bmj* 2016;352:i1547.
9. Davies BR, Howells S, Jenkins M. Early detection and treatment of postnatal depression in primary care. *Journal of Advanced Nursing* 2003;44(3):248-55.
10. Yonkers KA, Wisner KL, Stewart DE, et al. The management of depression during pregnancy: a report from the American Psychiatric Association and the American College of Obstetricians and Gynecologists. *General hospital psychiatry* 2009;31(5):403-13.
11. LaRocco-Cockburn A, Melville J, Bell M, et al. Depression screening attitudes and practices among obstetrician–gynecologists. *Obstetrics & Gynecology* 2003;101(5):892-98.
12. Bosanquet K, Bailey D, Gilbody S, et al. Diagnostic accuracy of the Whooley questions for the identification of depression: a diagnostic meta-analysis. *BMJ open* 2015;5(12):e008913.

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13. Sockol LE, Epperson CN, Barber JP. A meta-analysis of treatments for perinatal depression. *Clinical psychology review* 2011;31(5):839-49.
14. Rahman A, Fisher J, Bower P, et al. Interventions for common perinatal mental disorders in women in low-and middle-income countries: a systematic review and meta-analysis. *Bulletin of the World Health Organization* 2013;91:593-601I.
15. Dennis CL, Brown JVE, Brown HK. Interventions (other than psychosocial, psychological and pharmacological) for treating postpartum depression. *Cochrane Database of Systematic Reviews* 2019
16. Dennis CL, Dowswell T. Psychosocial and psychological interventions for preventing postpartum depression. *Cochrane database of systematic reviews* 2013(2)
17. O'Connor E, Senger CA, Henninger ML, et al. Interventions to prevent perinatal depression: evidence report and systematic review for the US Preventive Services Task Force. *Jama* 2019;321(6):588-601.
18. Yin J, Nisar A, Waqas A, et al. Psychosocial interventions on perinatal depression in China: a systematic review and meta-analysis. *Journal of Affective Disorders* 2020
19. Claridge AM. Efficacy of systemically oriented psychotherapies in the treatment of perinatal depression: a meta-analysis. *Archives of women's mental health* 2014;17(1):3-15.
20. Lever Taylor B, Cavanagh K, Strauss C. The effectiveness of mindfulness-based interventions in the perinatal period: a systematic review and meta-analysis. *PloS one* 2016;11(5):e0155720.
21. Cluxton-Keller F, Bruce ML. Clinical effectiveness of family therapeutic interventions in the prevention and treatment of perinatal depression: A systematic review and meta-analysis. *PLoS One* 2018;13(6):e0198730.
22. Nakamura A, van der Waerden J, Melchior M, et al. Physical activity during pregnancy and postpartum depression: Systematic review and meta-analysis. *Journal of affective disorders* 2019;246:29-41.
23. Ng QX, Venkatanarayanan N, Loke W, et al. A meta-analysis of the effectiveness of yoga-based interventions for maternal depression during pregnancy. *Complementary therapies in clinical practice* 2019;34:8-12.
24. Stephens S, Ford E, Paudyal P, et al. Effectiveness of psychological interventions for postnatal depression in primary care: a meta-analysis. *The Annals of Family Medicine* 2016;14(5):463-72.
25. Molyneaux E, Howard LM, McGeown HR, et al. Antidepressant treatment for postnatal depression. *Cochrane Database of Systematic Reviews* 2014(9)
26. Excellence NIfHaC. Antenatal and postnatal mental health: clinical management and service guidance (Clinical guidelines 192) 2020 [Available from: https://www.nice.org.uk/guidance/cg192 accessed 12 May 2020.
27. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS med* 2009;6(7):e1000097.
28. Hutton B, Salanti G, Caldwell DM, et al. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. *Annals of internal medicine* 2015;162(11):777-84.
29. Hirst KP, Moutier CY. Postpartum major depression. *American family physician* 2010;82(8):926-33.
30. Kuo C, Schonbrun YC, Zlotnick C, et al. A qualitative study of treatment needs among pregnant and postpartum women with substance use and depression. *Substance use & misuse* 2013;48(14):1498-508.
31. Sterne JA, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *bmj* 2019;366
32. Higgins JP, Thomas J, Chandler J, et al. Cochrane handbook for systematic reviews of interventions: John Wiley & Sons 2019.

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33. Dias S, Ades AE, Welton NJ, et al. Network meta-analysis for decision-making: John Wiley & Sons 2018.
34. Efthimiou O, Debray TP, van Valkenhoef G, et al. GetReal in network meta-analysis: a review of the methodology. *Research synthesis methods* 2016;7(3):236-63.
35. Schünemann H, Brożek J, Guyatt G, et al. GRADE handbook for grading quality of evidence and strength of recommendations. Updated October 2013. The GRADE Working Group, 2013. *Available from guidelinedevelopment org/handbook* 2013

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Author contribution:

RDS drafted the manuscript. KL is the principal investigator of the study and is responsible for conducting the study overall. The study's research question and design were conceived by RDS and KL. Search terms for all databases were constructed and pilot tested by RDS, JG, HLI and SCH. Screening forms, data extraction forms and risk of bias forms were developed and piloted by RDS, KL, JG, SCH and IPH. Assistance with preparation of the manuscript and clinical expertise into the design of the study's PICOS and inclusion/exclusion criteria was given by CAW. Development and plan of analyses were prepared by RDS, DF and SA. All authors contributed to the preparation of this manuscript.

Competing interests' statement.

All authors have no competing interests to declare.

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Table 1. PICOS for search strategy and study selection criteria

Objective 1	
Population	Perinatal mothers, or if not identified as perinatal within the study; females between 20th week of gestation to 4 weeks after birth, no limitation of setting, excluding those currently experiencing a depression episode
Intervention	Interventions that aimed to prevent perinatal depression including pharmacological and non-pharmacological interventions, interventions were not limited for setting
Comparison	Studies could were not limited for comparisons groups including other active interventions or non-active controls
Outcome	Depression diagnosis (determined through clinical interview)
Study design	Randomised controlled trials only
Objective 2	
Population	Perinatal mothers, or if not identified as perinatal within the study; females between 20th week of gestation to 4 weeks after birth, no limitation of setting, diagnosed or known to be currently experiencing a depression episode
Intervention	Interventions that aimed to treat perinatal depression symptoms including pharmacological and non-pharmacological interventions, interventions were not limited for setting
Comparison	Studies could were not limited for comparisons groups including other active interventions or non-active controls
Outcome	Measurements of depression severity or symptoms
Study design	Randomised controlled trials only

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Table 2. Inclusion and exclusion criteria for study selection

Inclusion Criteria	Exclusion Criteria
• Research paper in a peer-reviewed journal • Studies that met the PICOS criteria for either review question 1 or 2.	• Populations that include persons with substance abuse, psychotic or developmental disorders or medical conditions, in long term care, residential facilitates or those in institutions (psychiatric inpatients), studies that include subsets these populations were excluded if they exceeded 50% of the study sample • Study designs or than randomised controlled trials

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1. Perinatal Care/
2. Prenatal Care/
3. Pregnant Women/
4. Pregnan*.ti,ab.
5. perinatal.ti,ab.
6. prenatal.ti,ab.
7. 1 OR 2 OR 3 OR 4 OR 5 OR 6
8. exp Depressive Disorder/
9. Depression/
10. (depression or depressive or depressed).ti,ab.
11. (melancholi$ or dysphori$ or dysthymi$).ti,ab.
12. 8 or 9 or 10 or 11
13. Depression, Postpartum/
14. controlled clinical trial.pt.
15. randomized controlled trial.pt.
16. (randomi#ed or randomi#ation).ab,ti.
17. randomly.ab.
18. (random* adj3 (administ* or allocat* or assign* or class* or control* or determine* or divide* or distribut* or expose* or fashion or number* or place* or recruit* or subsitut* or treat*)).ab.
19. placebo*.ab,ti.
20. drug therapy.fs.
21. trial.ab,ti.
22. (control* adj3 (trial* or study or studies)).ab,ti.
23. ((singl* or doubl* or tripl* or trebl*) adj3 (blind* or mask* or dummy*)).mp.
24. clinical trial, phase ii/ or clinical trial, phase iii/ or clinical trial, phase iv/ or randomized controlled trial/ or pragmatic clinical trial/
25. (quasi adj (experimental or random$)).ti,ab.
26. ((waitlist* or wait* list* or treatment as usual or tau) adj3 (control or group)).ab,ti.
27. (14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26)
28. (7 and 12 and 13)
29. (27 and 28)
30. animal/ NOT human/
31. (29 NOT 30)

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Location
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Title page (page 1)
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A. This is an original protocol submission
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	Abstract (page 3)
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Title page (page 1)
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Author contribution (page 16)
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A. This is not an amendment
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Funding statement (page 16)
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Introduction (page 5-7)
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Introduction (page 7)
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting,	Eligibility section (page 8), Table 1, and Table 2

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		time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Sources of information section (page 8-9)
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Search strategy section (page 9) and Appendix 1
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Methods section (page 8-11)
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Data analysis section (page 11-13)
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Data analysis section (page 11-13) & data extraction section (page 8-10)
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Risk of bias assessment section (page 10-11)
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	Data analysis effect measurements section (10-11)
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Network meta-analysis implementation section (page 11-13)
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	Network meta-analysis implementation section (page 11-13)
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Network meta-analysis implementation section (page 11-13)
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication	Risk of bias assessment section (page 10-11)

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		bias across studies, selective reporting within studies)	
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Ethics and dissemination section (page 14)

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Protocol of a systematic review and network meta-analysis for the prevention and treatment of perinatal depression.

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Title: Protocol of a systematic review and network meta-analysis for the prevention and treatment of perinatal depression.

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Author contribution:

RDS drafted the manuscript. KL is the principal investigator of the study and is responsible for conducting the study overall. The study's research question and design were conceived by RDS and KL. Search terms for all databases were constructed and pilot tested by RDS, JG, IHL and SCH. Screening forms, data extraction forms and risk of bias forms were developed and piloted by RDS, KL, JG, SCH and IPH. Assistance with preparation of the manuscript and clinical expertise into the design of the study's PICOS and inclusion/exclusion criteria was given by CAW. Development and plan of analyses were prepared by RDS, DF and SA. All authors contributed to the preparation of this manuscript.

Keywords: perinatal depression, meta-analysis, systematic review

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Abstract

Introduction

Perinatal depression is common and can often lead to adverse health outcomes for mother and child. Multiple pharmacological and non-pharmacological treatments have been evaluated against usual care or placebo controls in meta-analyses for preventing and treating perinatal depression compared. It is not yet established which of these candidate treatments might be the optimal approach for prevention or treatment.

Methods and analysis

A systematic review and Bayesian network meta-analyses will be conducted. Eight electronic databases shall be searched for randomised controlled trials that have evaluated the effectiveness of treatments for prevention and/or treatment of perinatal depression. Screening of articles shall be conducted by two reviewers independently. One network meta-analysis shall evaluate the effectiveness of interventions in preventing depression during the perinatal period. A second network meta-analysis shall compare the effectiveness of treatments for depression symptoms in women with perinatal depression. Bayesian 95% credible intervals shall be used to estimate the pooled mean effect size of each treatment, and surface under cumulative ranking area (SCURA) will be used to rank the treatments' effectiveness.

Ethics and dissemination

We shall report our findings so that healthcare providers can make informed decisions on what might be the optimal approach for addressing perinatal depression to prevent cases and

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improve outcomes in those suffering with depression through knowledge exchange workshops, international conference presentation, and journal article publication.

Prospero registration number: Prospero ID number CRD42020200081

Strengths and limitations of this study

- This planned systematic review and network meta-analysis shall evaluate all available evidence from randomised controlled trials to evaluate the comparative effectiveness of each intervention
- This study shall be conducted following the latest guidelines of the Cochrane Handbook for systematic reviews of interventions
- Heterogeneity shall be assessed in the network meta-analysis model within the direct-comparisons model and comparing consistency between the direct and indirect model
- A limitation of this approach could be the different contexts of managing perinatal depression across studies in different regions and cultures
- To minimize the impact of this subgroup analysis shall be conducted grouping studies by region, allowing for comparisons of interventions within different regions

Introduction

Background

Depression is the leading cause of disability worldwide, it is a major contributor to the global burden of disease, affecting a variety of populations[1]. Depression experienced during pregnancy and after birth, also known as perinatal depression, is common and can affect up to 20% of mothers[2]. Previous systematic reviews have shown that the prevalence of perinatal

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depression is generally higher in low-to-middle-income countries than high-income countries in both the antenatal and postnatal stages [3 4]. With perinatal mental disorders, including depression, being more prevalent in mothers who are the most socioeconomically disadvantaged [4]. Cultural factors also have been indicated to be sources of inequality for perinatal mental illness, these include cultural gender-bias, gender-biased violence; both physically and mentally [4].

Perinatal depression can cause a range of adverse health outcomes for women and the development of their children. Depression during pregnancy can lead to multiple problems, including premature delivery, gastrointestinal pain, poorer self-report health and functioning, it can also lead to an increased risk of smoking or alcohol abuse[5 6]. Longer-term depression beyond one year postpartum, can lead to later problems during parenting, including lower interaction and sensitivity between mother and infant. Long lasting depression has been shown to lead to further difficulties in later years for the offspring, including emotional and behavioural difficulties[7-9]. Successful prevention of postnatal depression occurring can be achieved. Identifying those with depression early; either during the antenatal period (during pregnancy) or in the postnatal period (up to 1 year postpartum) provides a critical opportunity for earlier treatments and prevents poorer outcomes from occurring [10 11].

Despite its significant burden on maternal and child health, less than half of pregnant women suffering from depression are identified within healthcare[12]. Attitudes towards identifying cases of perinatal depression among clinicians are positive. Still, there is a need for support strategies that can identify and treat those at risk of perinatal depression within routine practice[13]. A systematic review suggested that the Whooley questions, a set of 2-item yes/no

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answered questions were a valid and feasible approach for identifying possible positive cases of perinatal depression[14].

Once identified, healthcare services can provide interventions for preventing those at risk of depression occurring in the future or offering treatments for those with depression. There is a wide variety of interventions that are shown to be effective in treating depression symptoms in perinatal women compared to non-active controls: psychological interventions, pharmacological interventions or combinations of both[15]; psychoeducation or parenting education[16]; psychosocial interventions for treatment and prevention[17-20]; systemically oriented psychotherapies[21]; mindfulness[22]; family therapy[23]; physical activity[24]; and yoga-based interventions[25]. In later stage postnatal women, meta-analyses have suggested cognitive behavioural therapy, interpersonal therapy, counselling and other psychological interventions are effective in treating depression symptoms when compared to usual care[26]. Another meta-analysis on antidepressants for postnatal depression in a small number of studies show that selective serotonin reuptake inhibitors (SSRI) are effective for depression compared to placebo[27].

Clinical guidelines recommend screening and treatment for perinatal depression, these guidelines do not provide recommendations on which treatments are most effective[28]. Treatment options for depression during pregnancy may vary depending on different severities of depression[10]. Many treatments previously evaluated were identified as effective on depression symptoms compared to usual care or placebo, but the relative comparability of these treatment has not previously been investigated[26 29]. Relative comparisons of different treatments would allow healthcare providers to make informed decisions on how different active treatments can be compared. The relative comparison of treatments could

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also provide evidence for the optimal approach to treating perinatal depression based on all available evidence.

Rationale

Using a Bayesian network meta-analysis facilitates all interventions to be compared equally with one-another by using the direct evidence (within study comparisons of treatments) and indirect evidence (comparing treatments across different studies), which previous systematic review studies and clinical guidelines in perinatal depression have not yet explored. This approach can provide evidence for the relative comparative effectiveness of each treatment and potentially identify the optimal approach for preventing cases and treating symptoms of perinatal depression.

We will conduct a comprehensive systematic review of available peer reviewed published trial studies for all pharmacological and non-pharmacological interventions, addressing perinatal depression by conducting a network meta-analysis. Based on this we will be able to compare each treatments' effectiveness with one-another and recommend types of interventions that may optimally address the prevention and treatment of perinatal depression. An example of this could be making relative comparisons on the effectiveness of interventions that require fewer resources for health providers to implement that is scalable against more resource intensive interventions that require trained specialists or equipment to implement and the level of trade off in clinical effectiveness between those interventions. Another advantage of Bayesian network meta-analysis is statistical certainty can be estimated, this allows for identifying potentially promising interventions with low levels of statistical certainty that may require further investigation to establish effectiveness.

Objectives

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This study will assess the clinical benefits of different interventions for addressing the prevention and treatment of perinatal depression. Two objectives have been developed for this study.

1.) To identify the optimal approach for preventing perinatal depression in women.

2.) To identify the optimal approach for the treatment of perinatal depression in women.

Methods

The protocol of this study has been developed following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) guidelines (**appendix 1**) and the PRISMA extension statement on the reporting of systematic reviews that incorporate network meta-analysis of health care interventions[30 31].

Eligibility criteria

Study selection and eligibility criteria were based on the PICOS related to objectives 1 and 2 (**table 1.**). Briefly, studies on participants who are perinatal women between 20 weeks gestation to 1 year after birth will be included. We selected 1 year after birth to reflect the time period in which there is risk of postpartum depression occurring between day 1 to 1 year postpartum[32]. We did not specify any limitations on interventions as this review aims to identify and evaluate all intervention types that address depression within the target population. Given the advantages of the network meta-analysis approach, no limitations will be placed on the comparison group for studies. Outcomes for Objective 1 will be: confirmed cases of depression and for Objective 2: measurements of depression severity or symptoms. Study design will only include randomised controlled trials to minimise the risk of bias when comparing effectiveness of interventions. Eligibility is displayed in table 2. Studies that include

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participants with substance abuse, psychotic or developmental disorders or medical conditions, long-term care, residential facilities or those in institutions (psychiatric inpatients) were excluded as the treatment needs of these populations' depression symptoms differ compared to those with depression alone[33]. Included articles were limited to those written in English, there was no limitation of publication year for included articles.

Table 1. PICOS for search strategy and study selection criteria

Objective 1	
Population	Perinatal mothers, or if not identified as perinatal within the study; females between 20th week of gestation to 1 year after birth, no limitation of setting, excluding those currently experiencing a depression episode
Intervention	Interventions that aimed to prevent perinatal depression including pharmacological and non-pharmacological interventions, interventions were not limited for setting
Comparison	Studies could were not limited for comparisons groups including other active interventions or non-active controls
Outcome	Depression diagnosis (determined through clinical interview)
Study design	Randomised controlled trials only
Objective 2	
Population	Perinatal mothers, or if not identified as perinatal within the study; females between 20th week of gestation to 1 year after birth, no limitation of setting, diagnosed or known to be currently experiencing a depression episode
Intervention	Interventions that aimed to treat perinatal depression symptoms including pharmacological and non-pharmacological interventions, interventions were not limited for setting
Comparison	Studies could were not limited for comparisons groups including other active interventions or non-active controls
Outcome	Measurements of depression severity or symptoms
Study design	Randomised controlled trials only

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Sources of information

Databases searched electronically will be MEDLINE, British Nursing Index (BNI), EMBASE, Cumulative Index to Nursing & Allied Health (CINAHL Plus), PsycINFO, Cochrane Central Register of Controlled Trials (CENTRAL), Cochrane Database of Systematic Reviews (CDSR) and Web of Science (WoS). We selected MEDLINE, EMBASE and WoS databases based on recommendations for Cochrane Handbook covering major health sciences topics. We also selected specialist subject databases BNI, CINAHL Plus and PsycINFO based on their relevance to the study objectives. World Health Organization's trials portal and clinicaltrials.gov will be searched to identify unpublished studies or studies still ongoing.

We shall conduct searches of reference lists and forward citation of identified and included studies using the Web of Science database for additional papers. We shall exclude studies that are published systematic review or literature review identified during our electronic searches but shall examine the reference lists for additional candidate studies.

Search Strategy

Searches of online databases will commence in August 2021. The search strategy has been developed based on the two sets of PICOS with one for each review question. We identified all search terms, related to the two sets of PICOS, from previously published meta-analyses on the prevention or treatment of perinatal depression[15-25] to maximise sensitivity of our search strategy in identifying all definitions of perinatal depression and minimise risk to missing relevant studies. An example full list of search terms for the MEDLINE database can be found in **appendix 2**.

Data management

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Studies retrieved in our search strategy shall be downloaded and stored in EndNote (X9) where duplicates across difference sources will be removed.

Study selection process

Screening of studies shall be conducted using the inclusion and exclusion criteria (**table 2)**. Two reviewers (RS, JG, SGH or IHP) shall conduct study selection, with both reviewers reviewing each retrieved study independently. Differences in assessment or any disagreements over the eligibility of studies were resolved by discussion, and in cases of disagreement, the third reviewer (KL) would be consulted. Screening of studies shall be conducted in two different stages: 1.) title and abstract, where studies will only be excluded if there is a clear disparity to eligibility criteria, if it is unclear, then articles or included articles will be further screened in the stage; 2.) full article screening, where eligibility can be decided based on all reported article information, including supplementary materials. The rationale for the exclusion of studies will be recorded. Piloting of the data selection process shall take place prior to the full study selection, with 100 articles randomly selected from the search retrieved. Adjustments to the inclusion and exclusion criteria and study selection process may be made following piloting.

Table 2. Inclusion and exclusion criteria for study selection

Inclusion Criteria	Exclusion Criteria
• Research paper in a peer-reviewed journal • Studies that met the PICOS criteria for either review question 1 or 2.	• Populations that include persons with substance abuse, psychotic or developmental disorders or medical conditions, in long term care, residential facilitates or those in institutions (psychiatric inpatients), studies that include subsets these populations were excluded if they exceeded 50% of the study sample • Study designs or than randomised controlled trials

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Data extraction

Data extraction shall be conducted by two of the reviewers (RS, JG, SGH or IHP) independently using a standardised extraction form. Differences in assessment or any disagreements for data extraction of studies will be resolved by discussion, and in cases of disagreement, the third reviewer (KL) will be consulted. The data extraction forms were taken from the Cochrane Consumers and Communication Review Group's Data Extraction Template for Cochrane Reviews, and were modified to fit this systematic review. The extracted information will include study setting, study participant demographics and baseline characteristics, details of the intervention and control conditions, study methodology, recruitment and study completion rates, outcomes and times of measurement, indicators of acceptability to users, suggested mechanisms of intervention action, and information for the assessment of the risk of bias. Missing data will be requested from study authors.

Risk of bias assessment

Risk of bias will be assessed using version 2 of the Cochrane Risk of Bias tool (RoB2)[34]. This tool includes 5-domains based on the risk of biases within the randomisation process, deviation from intervention design, missing outcome data, measurement of outcome, and reporting of results. The risk of bias assessment shall be completed following the guidance released with the RoB2. Possible risk of bias judgments shall include 1) low risk, 2) some concerns and 3) high risk of bias. Results of the risk of bias shall be presented within each domain of risk of bias, as well as an overall score for each study. Overall risk of bias shall be judged based on the suggested criteria of the RoB2 guidance document. Piloting and adaption of the wording in the risk of bias assessment, if necessary, shall be conducted prior to the full review. Risk of bias will be evaluated by two of the reviewers (RS, JG, SGH or IHP)

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independently evaluate the risk of bias. Differences in assessment or any disagreements will be resolved by discussion, and in cases of disagreement, the third reviewer (KL) will be consulted.

Data analysis

Effect measurements

Data from each study shall be extracted with effect size calculated. For studies addressing review question 1, follow up data on the number of positive depressions cases in each arm shall be extracted allowing for relative risk (RR) to be calculated in each arms' comparison, with RR of less than 1 representing reduced risk of depression. For review question 2, treatment effect shall be calculated using mean difference (MD) if possible, or standardised mean difference (SMD) for depression severity. To calculate SMD; difference in changes (from baseline to follow-up) for intervention arms shall be used, divided by the pooled standard deviation (SD) of change. For studies with three or more arms a reference group shall be taken to calculate the SMD. Studies with negative SMD effect sizes representing improvements in reducing depression severity. Score changes will be used to control for possible baseline differences between study arms. For studies with multiple follow-up time points, we shall use the longest duration, up to a maximum of 1 year from the end of the intervention. In studies using median and interquartile ranges we shall impute these following the Cochrane Handbook [35]. Studies that do not report the SD of change from baseline will be imputed following the Cochrane Handbook. A correlation coefficient for imputation of SD of change shall be estimated based on the mean correlation in studies that do report all relevant data. If no studies report baseline, follow-up, and change values, we shall take the conservative value of r=0.5 to estimate SD of change. Where possible, we shall use the intention to treat

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sample for analyses. Interventions will be grouped for the network analysis using categories used by previous individual meta-analysis[16-25] as a framework, new emerging interventions not previously evaluated in meta-analysis shall be organised and grouped by agreement with the reviewing team.

Network meta-analysis implementation

Two network meta-analyses shall be conducted for depression prevention, using RR and treatment of depression severity, using SMD. We shall estimate model consistency by comparing the RR or SMD of the direct (within study comparisons) and indirect comparisons (between study comparisons), where direct comparisons are possible. The network meta-analysis will be conducted with a Bayesian Markov chain Monte Carlo (MCMC) method fitted using the Just Another Gibbs Samplers (JAGS) software within the R Statistical Software conducted within the BUGSnet package (R Core Team. 2020). We shall run four MCMC chains simultaneously in our model and construct two separate MCMC simulations to compare convergence. The Bayesian model shall run 5000 burn-in iterations and 100,000 simulation iterations. Convergence shall be assessed using the potential scale reduction factor (PSRF), where we expect the model to reduce to below 1.05. Heterogeneity (direct evidence) and model consistency (direct versus indirect) will be assessed using the node split function of the BUGSsnet package; sources of heterogeneity will be explored between studies. All results of each possible comparison of interventions shall be made using RR or SMD and 95% credible intervals (Crls), which can be considered Bayesian equivalent of confidence intervals. Rank probabilities will represent the probability of the ranking performance of each intervention type. Surface under the cumulative ranking (SUCRA) score will also be used to estimate the likelihood of the most effective intervention[36 37]. A limitation of the network meta-analysis

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approach is that treatments that have not been previously combined in trials cannot be combined in the meta-analysis. This precludes an investigation of whether or not combining two or more treatments provides any extra benefits than one treatment only.

Treatments for women with depression during or after pregnancy can vary in different populations and across disease severity[10]. To address this, network meta-regression and subgroup analysis shall be conducted to evaluate study characteristics that may influence the effect sizes of interventions within the network. Factors for exploring in the meta-regressions shall include the year of publication, the geographical region the study was conducted, study sample age, whether the study participants were at antenatal or early postnatal stage of motherhood, study sample's baseline depression severity (if possible), risk of bias in all 5 domains and overall risk of bias. Publication bias shall be assessed using two funnel plots of all included studies for each objective; trim and fill analysis shall also be conducted.

Patient involvement

Patients and the public were not involved in the design of the protocol or analysis of this study.

Ethics and dissemination

Ethics is not required for this study, given that this is a protocol for a systematic review, which utilises published data. The results of the review would be widely disseminated locally, nationally, and internationally. A paper would be submitted to a leading peer-review journal in this field, reporting of the study will adhere to the PRISMA extension statement on the reporting systematic reviews that incorporate network meta-analysis of health care interventions[31]. When presenting our findings from this study, the GRADE scoring for

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evidence and strength of recommendations shall be made following the criteria in the GRADE handbook[38]. The findings shall also be presented at a relevant international conference.

Reference List

1. Friedrich M. Depression is the leading cause of disability around the world. *JAMA* 2017;317(15):1517-17.
2. Woody C, Ferrari A, Siskind D, et al. A systematic review and meta-regression of the prevalence and incidence of perinatal depression. *Journal of affective disorders* 2017;219:86-92.
3. Shorey S, Chee CYI, Ng ED, et al. Prevalence and incidence of postpartum depression among healthy mothers: a systematic review and meta-analysis. *Journal of psychiatric research* 2018;104:235-48.
4. Fisher J, Mello MCd, Patel V, et al. Prevalence and determinants of common perinatal mental disorders in women in low-and lower-middle-income countries: a systematic review. *Bulletin of the World Health Organization* 2012;90:139-49.
5. Marcus SM. Depression during pregnancy: rates, risks and consequences. *Journal of Population Therapeutics and Clinical Pharmacology* 2009;16(1)
6. Grigoriadis S, VonderPorten EH, Mamisashvili L, et al. The impact of maternal depression during pregnancy on perinatal outcomes: a systematic review and meta-analysis. *The Journal of clinical psychiatry* 2013;74(4):321-41.
7. Murray L, Arteche A, Fearon P, et al. Maternal postnatal depression and the development of depression in offspring up to 16 years of age. *Journal of the American Academy of Child & Adolescent Psychiatry* 2011;50(5):460-70.
8. Velders FP, Dieleman G, Henrichs J, et al. Prenatal and postnatal psychological symptoms of parents and family functioning: the impact on child emotional and behavioural problems. *European child & adolescent psychiatry* 2011;20(7):341-50.
9. Kersten-Alvarez LE, Hosman CM, Riksen-Walraven JM, et al. Early school outcomes for children of postpartum depressed mothers: comparison with a community sample. *Child Psychiatry & Human Development* 2012;43(2):201-18.
10. Vigod SN, Wilson CA, Howard LM. Depression in pregnancy. *Bmj* 2016;352:i1547.
11. Davies BR, Howells S, Jenkins M. Early detection and treatment of postnatal depression in primary care. *Journal of Advanced Nursing* 2003;44(3):248-55.
12. Yonkers KA, Wisner KL, Stewart DE, et al. The management of depression during pregnancy: a report from the American Psychiatric Association and the American College of Obstetricians and Gynecologists. *General hospital psychiatry* 2009;31(5):403-13.
13. LaRocco-Cockburn A, Melville J, Bell M, et al. Depression screening attitudes and practices among obstetrician–gynecologists. *Obstetrics & Gynecology* 2003;101(5):892-98.
14. Bosanquet K, Bailey D, Gilbody S, et al. Diagnostic accuracy of the Whooley questions for the identification of depression: a diagnostic meta-analysis. *BMJ open* 2015;5(12):e008913.
15. Sockol LE, Epperson CN, Barber JP. A meta-analysis of treatments for perinatal depression. *Clinical psychology review* 2011;31(5):839-49.
16. Rahman A, Fisher J, Bower P, et al. Interventions for common perinatal mental disorders in women in low-and middle-income countries: a systematic review and meta-analysis. *Bulletin of the World Health Organization* 2013;91:593-601I.
17. Dennis CL, Brown JVE, Brown HK. Interventions (other than psychosocial, psychological and pharmacological) for treating postpartum depression. *Cochrane Database of Systematic Reviews* 2019
18. Dennis CL, Dowswell T. Psychosocial and psychological interventions for preventing postpartum depression. *Cochrane database of systematic reviews* 2013(2)

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19. O'Connor E, Senger CA, Henninger ML, et al. Interventions to prevent perinatal depression: evidence report and systematic review for the US Preventive Services Task Force. *JAMA* 2019;321(6):588-601.
20. Yin J, Nisar A, Waqas A, et al. Psychosocial interventions on perinatal depression in China: a systematic review and meta-analysis. *Journal of Affective Disorders* 2020
21. Claridge AM. Efficacy of systemically oriented psychotherapies in the treatment of perinatal depression: a meta-analysis. *Archives of women's mental health* 2014;17(1):3-15.
22. Lever Taylor B, Cavanagh K, Strauss C. The effectiveness of mindfulness-based interventions in the perinatal period: a systematic review and meta-analysis. *PloS one* 2016;11(5):e0155720.
23. Cluxton-Keller F, Bruce ML. Clinical effectiveness of family therapeutic interventions in the prevention and treatment of perinatal depression: A systematic review and meta-analysis. *PLoS One* 2018;13(6):e0198730.
24. Nakamura A, van der Waerden J, Melchior M, et al. Physical activity during pregnancy and postpartum depression: Systematic review and meta-analysis. *Journal of affective disorders* 2019;246:29-41.
25. Ng QX, Venkatanarayanan N, Loke W, et al. A meta-analysis of the effectiveness of yoga-based interventions for maternal depression during pregnancy. *Complementary therapies in clinical practice* 2019;34:8-12.
26. Stephens S, Ford E, Paudyal P, et al. Effectiveness of psychological interventions for postnatal depression in primary care: a meta-analysis. *The Annals of Family Medicine* 2016;14(5):463-72.
27. Molyneaux E, Howard LM, McGeown HR, et al. Antidepressant treatment for postnatal depression. *Cochrane Database of Systematic Reviews* 2014(9)
28. National Institute for Health and Care Excellence . Antenatal and postnatal mental health: clinical management and service guidance (Clinical guidelines 192) 2020 [Available from: https://www.nice.org.uk/guidance/cg192 accessed 12 May 2020].
29. Howard LM, Khalifeh H. Perinatal mental health: a review of progress and challenges. *World Psychiatry* 2020;19(3):313-27.
30. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS med* 2009;6(7):e1000097.
31. Hutton B, Salanti G, Caldwell DM, et al. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. *Annals of internal medicine* 2015;162(11):777-84.
32. Yim IS, Stapleton LRT, Guardino CM, et al. Biological and psychosocial predictors of postpartum depression: systematic review and call for integration. *Annual review of clinical psychology* 2015;11
33. Kuo C, Schonbrun YC, Zlotnick C, et al. A qualitative study of treatment needs among pregnant and postpartum women with substance use and depression. *Substance use & misuse* 2013;48(14):1498-508.
34. Sterne JA, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;366
35. Higgins JP, Thomas J, Chandler J, et al. Cochrane handbook for systematic reviews of interventions: John Wiley & Sons 2019.
36. Dias S, Ades AE, Welton NJ, et al. Network meta-analysis for decision-making: John Wiley & Sons 2018.
37. Efthimiou O, Debray TP, van Valkenhoef G, et al. GetReal in network meta analysis: a review of the methodology. *Research synthesis methods* 2016;7(3):236-63.
38. Schünemann H, Brożek J, Guyatt G, et al. GRADE handbook for grading quality of evidence and strength of recommendations. Updated October 2013. The GRADE Working Group, 2013. *Available from guidelinedevelopment org/handbook* 2013

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Competing interests' statement.

All authors have no competing interests to declare.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol

Section and topic	Item No	Checklist item	Location
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	Title page
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	This is an original protocol submission
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	This systematic review protocol has been submitted to PROSPERO, we are awaiting confirmation
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Title page
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	Title page
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	NA- not an amendment
Support:			
Sources	5a	Indicate sources of financial or other support for the review	Statement on funding and role of funding is provided at end of manuscript
Sponsor	5b	Provide name for the review funder and/or sponsor	
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	Introduction background section
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	Two objectives at end of introduction
METHODS			

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Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	Table 1
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	Sources of information section
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	Appendix 1
Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	Methods section
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	Data analysis section
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	Data analysis section & data extraction section
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	Risk of bias assessment section
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	Data analysis effect measurements section
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	Network meta-analysis implementation section
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	Network meta-analysis implementation section
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	Network meta-analysis implementation section

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Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	Funnel plot and trim and fill analysis
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	Ethics and dissemination section

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1. Perinatal Care/
2. Prenatal Care/
3. Pregnant Women/
4. Pregnan*.ti,ab.
5. perinatal.ti,ab.
6. prenatal.ti,ab.
7. 1 OR 2 OR 3 OR 4 OR 5 OR 6
8. exp Depressive Disorder/
9. Depression/
10. (depression or depressive or depressed).ti,ab.
11. (melancholi$ or dysphori$ or dysthymi$).ti,ab.
12. 8 or 9 or 10 or 11
13. Depression, Postpartum/
14. controlled clinical trial.pt.
15. randomized controlled trial.pt.
16. (randomi#ed or randomi#ation).ab,ti.
17. randomly.ab.
18. (random* adj3 (administ* or allocat* or assign* or class* or control* or determine* or divide* or distribut* or expose* or fashion or number* or place* or recruit* or subsitut* or treat*)).ab.
19. placebo*.ab,ti.
20. drug therapy.fs.
21. trial.ab,ti.
22. (control* adj3 (trial* or study or studies)).ab,ti.
23. ((singl* or doubl* or tripl* or trebl*) adj3 (blind* or mask* or dummy*)).mp.
24. clinical trial, phase ii/ or clinical trial, phase iii/ or clinical trial, phase iv/ or randomized controlled trial/ or pragmatic clinical trial/
25. (quasi adj (experimental or random$)).ti,ab.
26. ((waitlist* or wait* list* or treatment as usual or tau) adj3 (control or group)).ab,ti.
27. (14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26)
28. (7 and 12 and 13)
29. (27 and 28)
30. animal/ NOT human/
31. (29 NOT 30)

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